NEPHRA

A dialysis digital twin.

Anticipate a pressure fall.

Make the review easier to prepare.

Historical record + changing sensors + patient-specific simulation

Working proof of concept | October 2026

The problem: a pressure fall during dialysis

Fluid removal and circulatory refilling can become unbalanced.

Staff must interpret pressure trends, treatment observations and patient history together.

A current-value threshold does not show the next 30 minutes.

Clinical background: KDIGO Blood pressure and volume management in dialysis, 2020. See README for the direct source link.

A narrow outcome makes the twin testable

Condition: chronic kidney disease requiring maintenance haemodialysis.

Endpoint: any future simulated SBP below 90 mmHg within 30 minutes.

User: a nurse or nephrologist reviewing a dialysis unit.

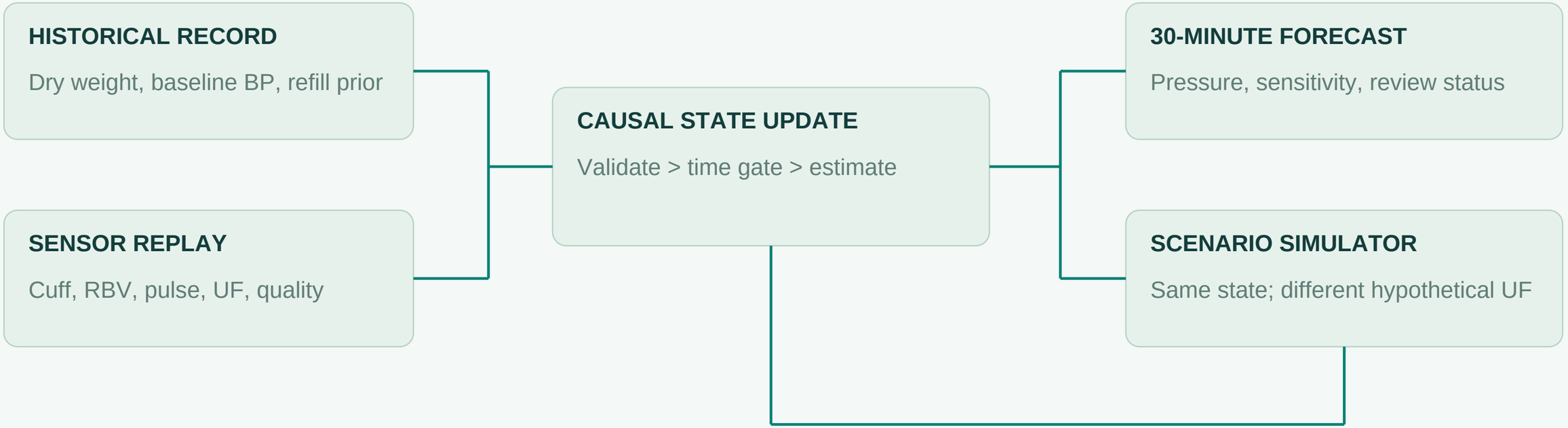
Two streams, one changing patient state

Historical priors: dry weight, baseline pressure, refilling and pressure sensitivity.

Dynamic observations: cuff pressure, relative blood volume, pulse and ultrafiltration.

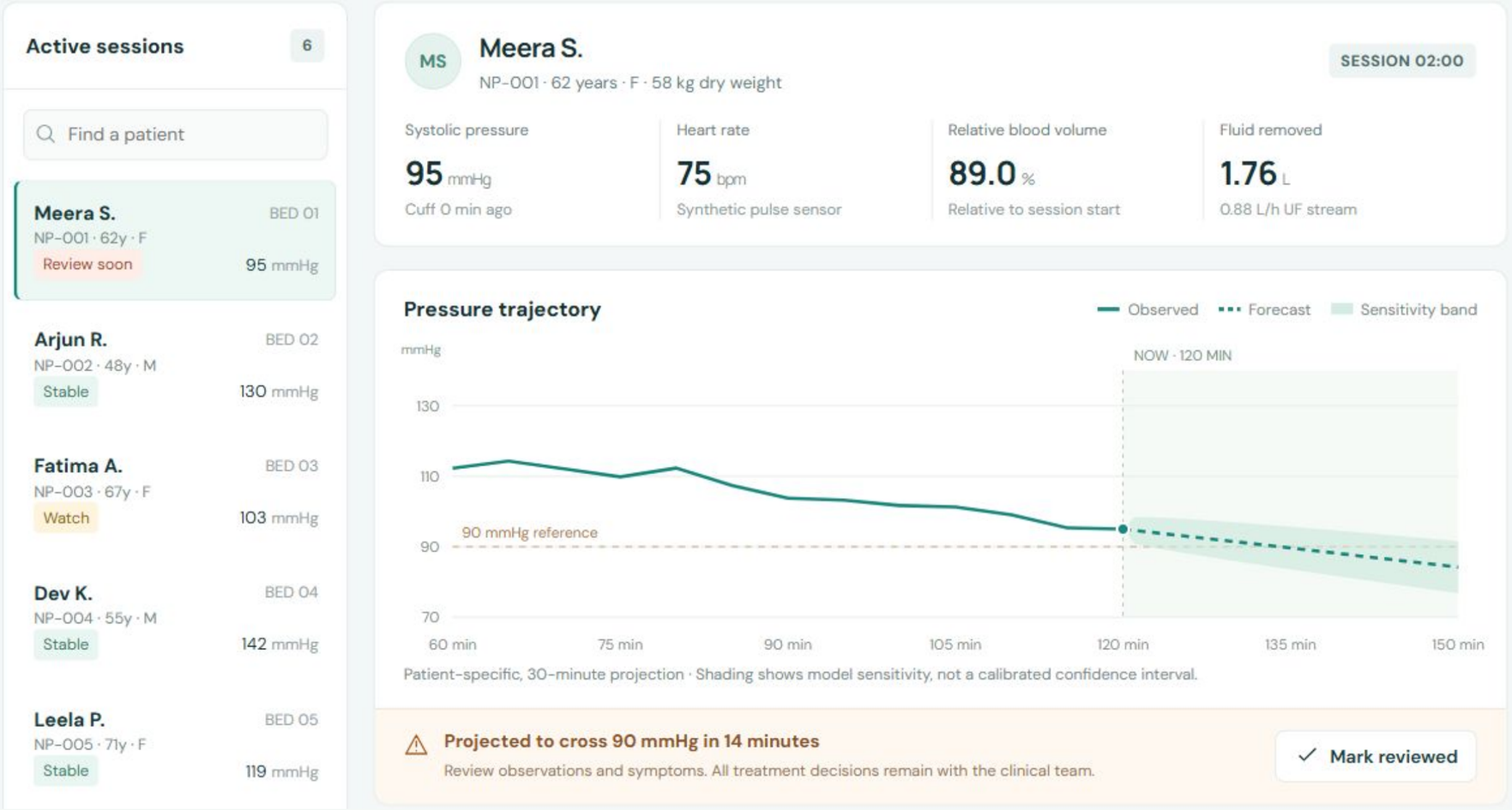
Five-minute cuffs; one-minute synthetic sensor observations.

The implemented architecture



All patient computation stays in the browser. No machine commands, hospital connection or clinical database.

The working dialysis command centre



Read the forecast, not just the latest number

The solid line is observed pressure; the dashed line is the model projection.

The reference line is 90 mmHg. A crossing is a review signal in this demo.

The shaded band is sensitivity analysis, not calibrated clinical confidence.

A mechanistic model with online correction

Volume change = fluid removed minus estimated refilling.

A scalar Kalman filter updates the deficit using relative blood volume.

The pressure projection combines physiology with the recent cuff trend.

Compare an isolated hypothetical scenario

Scenario inputs

Synthetic patient

Meera S. · NP-001

Hypothetical UF rate

0.62 L/hour

Observed rate: 0.88 L/hour

Comparison window

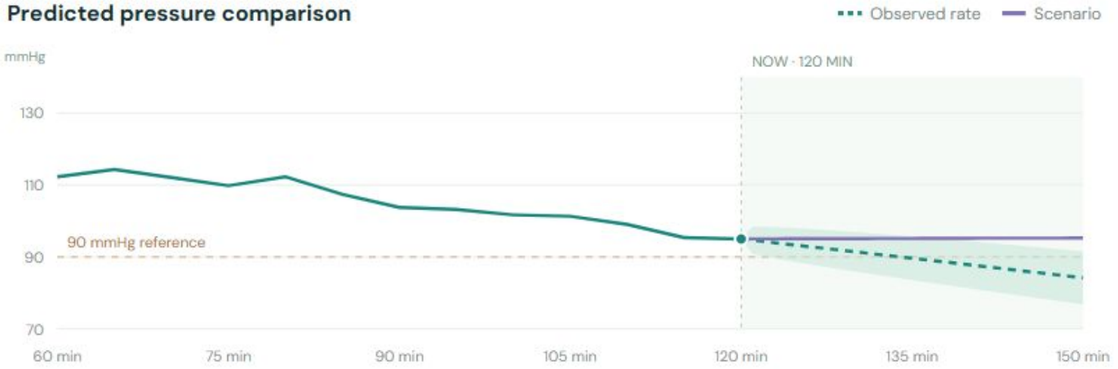
30 minutes

Both simulations start at minute 120 with the same estimated patient state.

Match observed rate

No saline, medication, or machine commands are issued. No scenario becomes the patient record.

Predicted pressure comparison



OBSERVED RATE PROJECTION

84.2 mmHg

Predicted lowest SBP

0.44 L removed over next 30 minutes

HYPOTHETICAL PROJECTION

95.0 mmHg

Predicted lowest SBP

0.31 L removed over next 30 minutes

The tradeoff stays visible

This scenario removes **0.13 L less** fluid over 30 minutes. The model estimates a **10.8 mmHg** change in the pressure nadir.

Pressure is only one part of dialysis care. Volume status, adequacy, symptoms and clinician judgement are outside this simplified simulation.

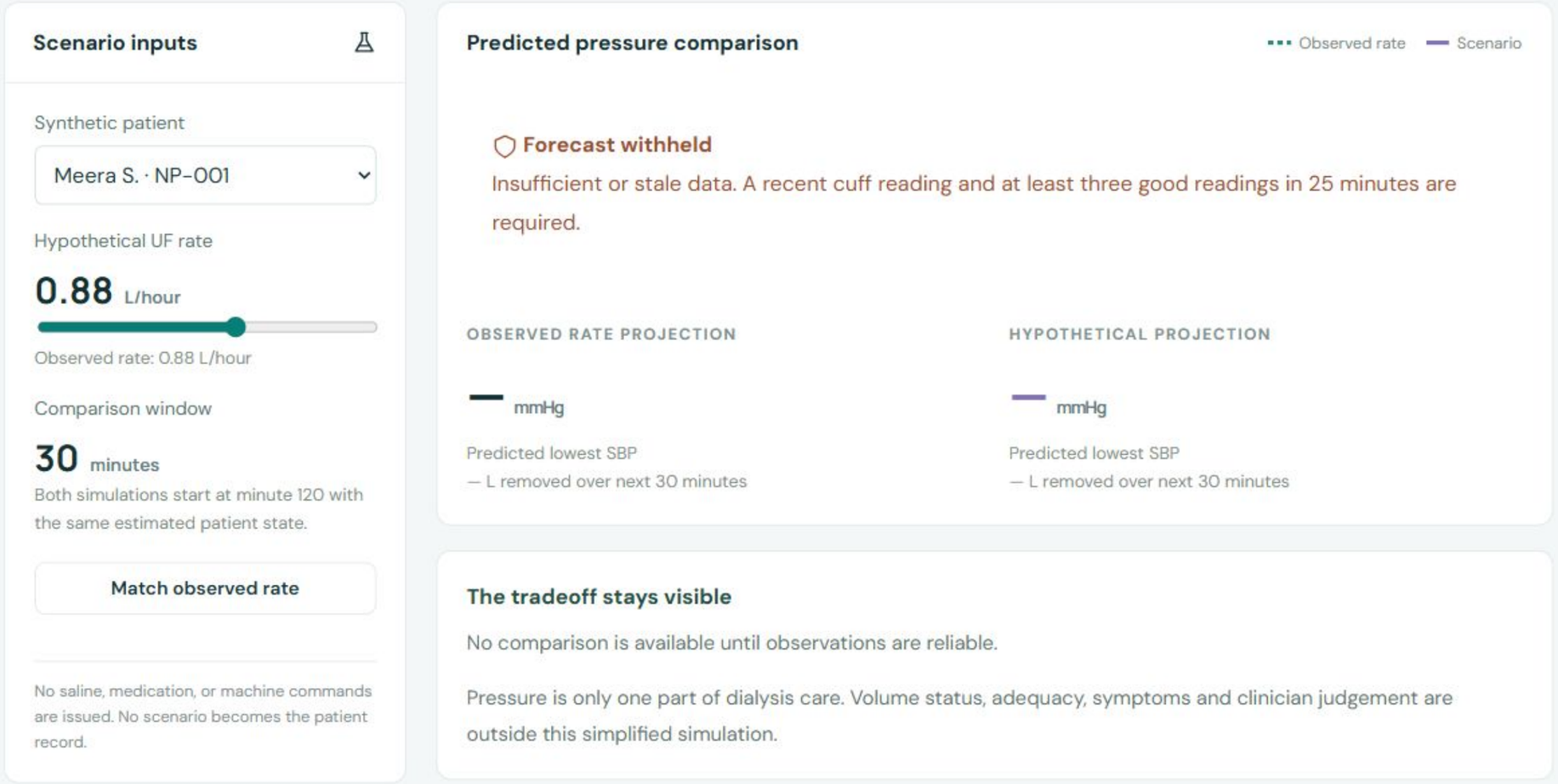
Every scenario exposes a tradeoff

A lower simulated removal rate may improve the projected pressure curve.

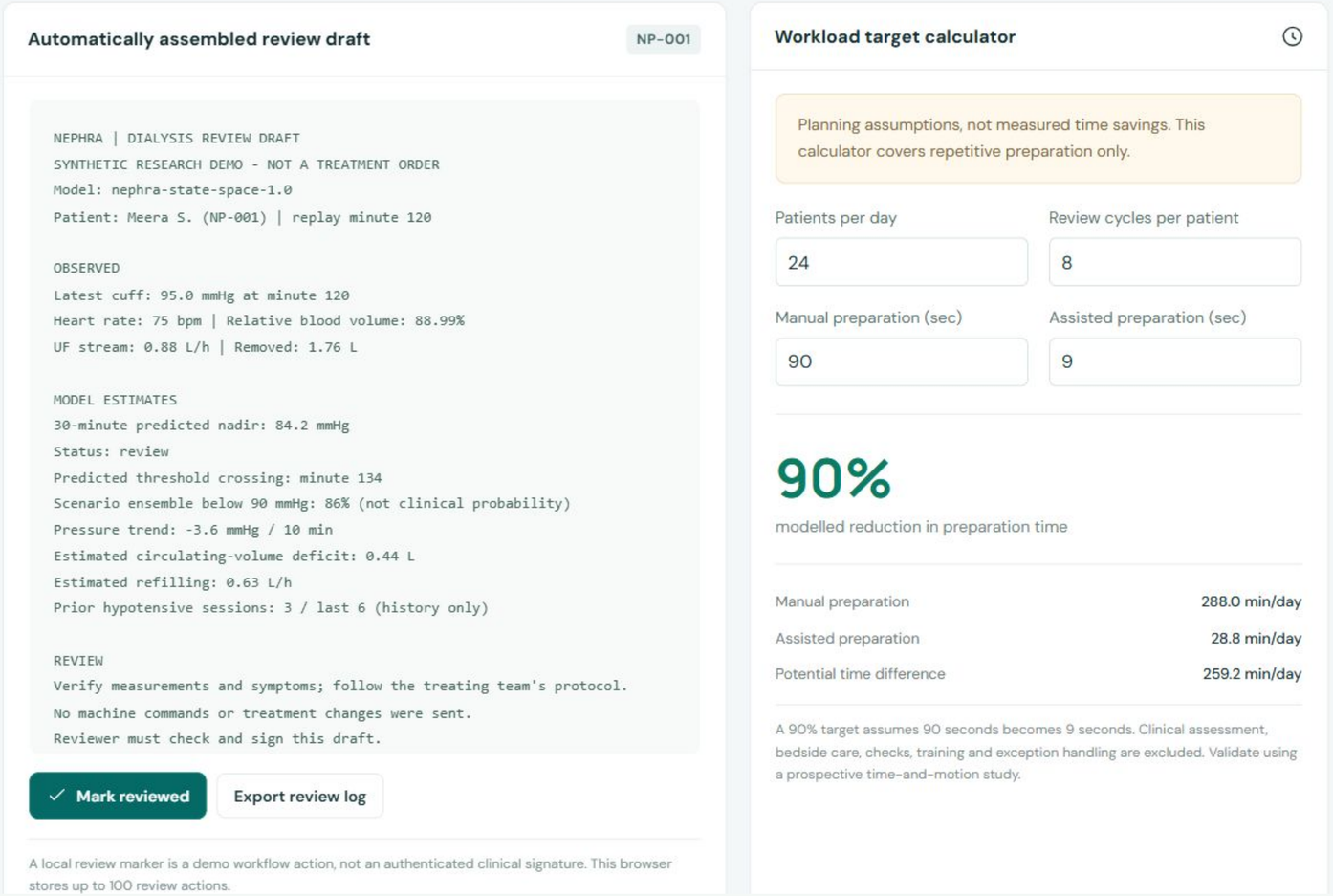
It also removes less fluid over the comparison window.

No treatment effect, recommended rate or clinical optimum is claimed.

When observations fail, the forecast stops



A review draft assembled from evidence



The 90% goal has an explicit denominator

90%

Target reduction in repetitive preparation time

90 seconds manual > 9 seconds assisted

24 patients x 8 review cycles

288 minutes > 28.8 minutes per day

Editable planning assumptions. No measured time savings; bedside care and clinical judgement are excluded.

Reproducible synthetic results

300 synthetic patients / 1,193 eligible windows



Mean absolute error of the next-30-minute pressure nadir (mmHg). Lower is better. Synthetic engineering evidence only.

Show the misses and the uncertainty

Twin: 111 of 128 future low-pressure windows detected; 17 missed.

Two false-positive windows in this synthetic benchmark.

No clinical calibration, independent-site validation or subgroup assurance.

Local computation and inspectable source

No API key, paid inference service or patient upload is required.

Synthetic JSON imports stay in browser memory.

Source, tests, generator, metrics and MIT license are included.

A clinical pilot must earn the next claim

Clinical partner and governed retrospective data.

Patient-separated evaluation, an external site and alarm-burden analysis.

Silent prospective observation before any assisted clinical workflow.

A funding case built around milestones

Initial users: dialysis nurses and nephrologists.

Product hypothesis: a read-only review layer for dialysis providers.

Fund discovery, independent validation and a governed pilot before scale.

Reproduce the prototype in three steps

Run npm start; open the local address printed by the server.

Run npm test to check model invariants and failure handling.

Run npm run evaluate to regenerate metrics and per-window predictions.

NEPHRA: a concrete, reviewable starting point

A working twin.

An explicit tradeoff.

A measurable workflow hypothesis.

A clear path to stronger evidence.

Team and college fields are intentionally left for the submitting team.